Lab 6: Rank and Column Space

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, October 14th, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Rank and Column Space (Chapter 5.3)

Suppose A is an $n \times d$ matrix. Then, $\text{rank}(A)$ is the number of linearly independent columns of A.

	Notation	**Description**	**Dimension**	**Subspace of**
Column space	$\text{colsp}(A)$	Span of the columns of A	$\text{rank}(A)$	$\mathbb{R}^n$
Row space	$\text{colsp}(A^T)$	Span of the rows of A	$\text{rank}(A)$	$\mathbb{R}^d$

CR Decomposition

Suppose A is an $n \times d$ matrix with rank r. A CR decomposition of A is a product of two matrices C and R, where $A = CR$ and:

- C is an $n \times r$ matrix and R is a $r \times d$ matrix
- C contains the linearly independent columns of A, selected left-to-right
- R tells how to "mix" the columns of C (which are linearly independent) to reconstruct the columns of A

Activity 1: Matrix–Vector Practice

Compute each product, or explain why it is undefined. For each defined product, state the shape of the output. Show your arithmetic.

a) $\begin{bmatrix} -8 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & 0 & 0 & 0 \\ 0 & 0 & 11 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ -4 \\ 2 \\ 6 \\ -3 \end{bmatrix}$

b) $\begin{bmatrix} 12 & -7 & \frac{1}{2} & 9 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 8 \\ -4 \end{bmatrix}$

c) $\begin{bmatrix} -5 & 8 & 3 & 12 \\ 7 & -2 & 6 & -9 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}$

Activity 2: Fundamentals and CR Decomposition

Let $X = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 & 4 \\ 2 & 5 & -2 & 7 & 11 & 10 \\ 4 & 8 & -4 & 12 & 16 & 16 \end{bmatrix}$.

a) Find a basis for colsp(X). What is rank(X)? Why? *Hint: Column 5 is a linear combination of columns 1 and 2. With this fact, you should be able to answer this relatively quickly.*

Let's keep working with $X = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 & 4 \\ 2 & 5 & -2 & 7 & 11 & 10 \\ 4 & 8 & -4 & 12 & 16 & 16 \end{bmatrix}$.

b) Fill in the blanks:

- colsp(X) is a _____-dimensional subspace of _____.
- colsp(X^T) is a _____-dimensional subspace of _____.

c) Find a CR decomposition of X. This shouldn't take very much work; review your work from part **a)** in finding a basis for colsp(X).

The key idea being assessed here is that in $A = CR$, the columns of C are linearly independent and a basis for colsp(A), while the rows of R are linearly independent and a basis for colsp(A^T)!

Activity 3: Outer Products

Recall that in lecture, we mentioned that you can use the transpose operation to write the dot product of two vectors in $\mathbb{R}^n$ as a matrix multiplication. For example, let

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \qquad \vec{v} = \begin{bmatrix} 2 \\ 0 \\ 4 \\ -1 \end{bmatrix}$$

Then,

$$\vec{v}^T\vec{u} = \begin{bmatrix} 2 & 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix} = 2(1) + 0(2) + 4(-1) + (-1)(3) = -5 = \vec{v} \cdot \vec{u}$$

A related operation that looks similar but does something very different is the **outer product** of two vectors: $\vec{u}\vec{v}^T$ instead of $\vec{v}^T\vec{u}$. For the same vectors,

$$\vec{u}\vec{v}^T = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 4 & -1 \\ 4 & 0 & 8 & -2 \\ -2 & 0 & -4 & 1 \\ 6 & 0 & 12 & -3 \end{bmatrix}$$

The dot product is a scalar; the outer product here is a 4×4 matrix.

Suppose $A = \vec{u}\vec{v}^T + \vec{w}\vec{z}^T$, where $\vec{u}, \vec{v}, \vec{w}, \vec{z} \in \mathbb{R}^n$ are non-zero vectors.

a) What is rank$(\vec{u}\vec{v}^T)$?

b) Under what conditions is rank$(A) = 2$? What about rank$(A) < 2$? *Hint: First, think about what happens when multiplying A by a vector $\vec{x} \in \mathbb{R}^n$. Can you write this as a linear combination of some other vectors? The case for rank$(A) = 2$ is more complicated than "$\vec{u}$ and $\vec{w}$ are linearly independent."*

Activity 4: Transforming a Triangle

The act of multiplying a vector by a matrix performs a **linear transformation** of the vector.

Suppose $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$. Multiplying $\vec{x}$ by the rotation matrix

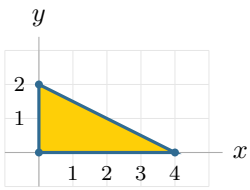
$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

gives

$$R_\theta \vec{x} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix},$$

which rotates $\vec{x}$ counterclockwise by θ about the origin.

Consider the triangle with vertices $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$.



a) Let $Q = R_{45^\circ}$. Write Q explicitly, then multiply all three original vertices by Q, showing your arithmetic. Draw the corresponding triangle and label its vertices.

b) Let $D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$. Since $D \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ y \end{bmatrix}$, it stretches a point horizontally by a factor of 2. Multiply all three original vertices by D, showing your arithmetic. Draw the corresponding triangle and label its vertices.

c) Compute QD. For each original vertex $\vec{x}$, multiply out $(QD)\vec{x}$. Draw the resulting triangle on a coordinate plane and label all three vertices. Explain why QD stretches first and then rotates.

d) Compute DQ. For each original vertex $\vec{x}$, multiply out $(DQ)\vec{x}$. Draw the resulting triangle on a second coordinate plane with the same scale and label all three vertices. Which operation happens first? How do the triangles for QD and DQ differ?

Note that the last page of this worksheet is empty if you need extra space to draw.

Activity 5: Can We Work Backward?

Let $A = \begin{bmatrix} 4 & -6 \\ -2 & 3 \end{bmatrix}$.

a) Find two different vectors $\vec{u}$ and $\vec{v}$ such that $A\vec{u} = A\vec{v} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$.

b) Is there a vector $\vec{x}$ such that $A\vec{x} = \begin{bmatrix} 12 \\ -5 \end{bmatrix}$? In your answer, reference colsp(A).

c) Using your $\vec{u}$ and $\vec{v}$ from part (a), find a nonzero vector $\vec{w}$ such that $A\vec{w} = \vec{0}$. Such a vector $\vec{w}$ is in the **null space** of A, nullsp(A). A's null space is the set of vectors that get sent to $\vec{0}$ when multiplied by A.